Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 06-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Phase 3b, Multi-center, Open-label, Treatment Optimization Study of Oral Asciminib in Patients With Chronic Myelogenous Leukemia in Chronic Phase (CML-CP) Previously Treated With 2 or More Tyrosine Kinase Inhibitors **Short Title/Acronym:** ASC4OPT **PostingID:** 261200053432658354 **Protocol Number:** CABL001A2302 **NCT Number:** NCT04948333 **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Sponsor / Company Name:** Novartis **Investigational Product:** Asciminib (Scemblix) **Phase of Development:** PHASE3 (Phase 3b) **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the Major Molecular Response (MMR) rate at 48 weeks in patients with CML-CP not in MMR at baseline who were previously treated with ≥2 tyrosine kinase inhibitors (TKIs) and are resistant or intolerant to their last TKI therapy. **Secondary Objectives:** To evaluate the MMR rates at weeks 12, 24, 36, 72, 96, and 144; time to MMR; early molecular responses; deep molecular responses (MR4.0 and MR4.5); cytogenetic responses; and patient-reported outcomes/ quality of life.

2.2 Background & Rationale Tyrosine kinase inhibitors (TKIs) are the standard of care for CML, but patients often experience resistance or intolerance to successive lines of TKI therapy. Asciminib is the first BCR::ABL1 inhibitor that works by Specifically Targeting the ABL Myristoyl Pocket (STAMP). This study aims to optimize the treatment of asciminib in adults with CML-CP (previously treated with ≥2 TKIs) by exploring two different dosing schedules (40 mg twice daily [BID] vs. 80 mg once daily [QD]) and potential dose escalation to maximize efficacy and safety.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (comparing two dosing schedules of the same investigational product) **Blinding:** Open-label **Randomization:** 1:1 **Phase ID / Requirement:** PHASE3 (Trial must be in PHASE3)

3.2 Planned Enrollment & Duration Targeted Enrollment: 199 (Original study design allocated approximately 186 patients: 156

patients resistant/intolerant to the last TKI and not in MMR at baseline; plus up to 30 additional patients intolerant to the last TKI but in MMR at baseline) **Number of Sites:** International, Multi-center **Estimated Study Start:** 01-Sep-2022 **Estimated Primary Completion:** Week 48 post-randomization of the final patient

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female patients with a diagnosis of CML-CP ≥ 18 years of age. 2. Treatment with a minimum of 2 or more prior TKIs (i.e., imatinib, nilotinib, dasatinib, bosutinib, radotinib, or ponatinib). 3. Warning or failure (adapted from the 2020 ELN Recommendations) or intolerance to the most recent TKI therapy at the time of screening. 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0, 1, or 2. 5. **Phase Requirement (Criteria Type: PHASE_REQUIREMENT):** Trial must be in PHASE3.

4.2 Exclusion Criteria 1. Known presence of the BCR::ABL1 T315I mutation (study specifically targets patients without this mutation). 2. Failure to meet adequate end organ function parameters as determined by central laboratory tests. 3. Previous treatment with asciminib.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Asciminib 80 mg daily, administered either as 40 mg twice daily (BID) or 80 mg once daily (QD). If a patient fails to achieve MMR at 48 weeks or loses response between weeks 48 and 108, the dose may be escalated to 200 mg QD, provided there is no prohibitive toxicity. **Route of Administration**: Oral **Duration of Treatment:** Up to 144 weeks of continuous treatment, followed by a 4-week post-treatment safety follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for CML as per protocol guidelines. Symptomatic treatments and general medications are permitted as necessary (e.g., mucolytics/expectorants such as **guaifenesin** [ATC Code: R05CA03] for cough/respiratory symptom relief). **Prohibited:** Other investigational drugs or unapproved concurrent antineoplastic therapies for CML.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Hematologic/Organ Function Labs, BCR::ABL1 baseline
Baseline	Day 0	Randomization (1:1), First Dose of Asciminib
Interim	Weeks 12, 24, 36, 48, 72, 96	Safety

Labs, Efficacy Assessment (BCR::ABL1 transcripts for MMR evaluation), Dose modification/escalation evaluation | | **End of Study** | Week 144 | Final Efficacy/Safety Assessments, IP Accountability, Transition to 4-week safety follow-up |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Major Molecular Response (MMR) rate at Week 48. MMR is defined as achieving a BCR::ABL1/ABL1 ratio of $\le 0.1\%$ on the international scale (IS). **Measurement Method:** Central Laboratory Analysis of BCR::ABL1 transcripts.

7.2 Secondary & Exploratory Endpoints 1. MMR rates at weeks 12, 24, 36, 72, 96, and 144. 2. Time to first documented MMR. 3. Rates of deep molecular responses (MR4.0 [BCR::ABL1 $\le 0.01\%$] and MR4.5 [BCR::ABL1 $\le 0.0032\%$]) up to 144 weeks. 4. Patient-reported outcomes and quality of life.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation of clinical and laboratory safety data. Dose escalations are strictly prohibited in the presence of Grade 3/4 toxicity or persistent Grade 2 toxicity unresponsive to optimal management. **Serious Adverse Events (SAEs):** Routine reporting to sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent safety review board governing dose escalation and trial progression.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) primary efficacy analysis will focus on the patients who are resistant/intolerant to their last TKI and *not* in MMR at baseline. The cohort of up to 30 patients in MMR at baseline will be analyzed separately. **Sample Size Justification:** Targeted Enrollment of 199 provides adequate power to assess the primary endpoint (MMR at 48 weeks) across the two randomized dosing schedules. **Handling of Missing Data:** Standard survival and response imputation conventions; missing molecular data at key milestones without documented response are typically treated as non-responses.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence and safety compliance. **Audit Trail:**

Standardized eCRF data entry, central laboratory molecular transcript tracking, and data clarification forms.

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11. Study Identifiers & Governance

Primary ID: CABL001A2302 **Registry Number:** NCT04948333 **PostingID:** 261200053432658354 **Sponsor/Lead Organization (Company):** Novartis **Study Title:** Asciminib Treatment Optimization in $\ge$ 3rd Line CML-CP **Official Regulatory Title:** A Phase 3b, Multi-center, Open-label, Treatment Optimization Study of Oral Asciminib in Patients With Chronic Myelogenous Leukemia in Chronic Phase (CML-CP) Previously Treated With 2 or More Tyrosine Kinase Inhibitors **Source Level:** 5

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12. Clinical Framework (CML Specific)

Disease Areas: Chronic Myelogenous Leukemia **Primary Condition / Name:** CML **Preferred Name:** Philadelphia chromosome positive chronic myelogenous leukemia **Preferred UMLS Name:** Chronic Myelogenous Leukemia **Definition:** A chronic myeloid leukemia characterized by the t(9;22)(q34;q11) chromosomal translocation, resulting in the presence of the Philadelphia chromosome and the BCR-ABL1 fusion gene. **Semantic Type:** Neoplastic Process **Taxonomy Codes:** * **MeSH Code:** D015464 * **HPO Codes:** HP:0005506 * **SNOMED ID:** 92818009 **Medical Methodology:** BCR::ABL1 Tyrosine Kinase Inhibitor (TKI) via ABL Myristoyl Pocket (STAMP) targeting **Optimization Target:** Major Molecular Response (MMR) definition of BCR::ABL1 $\le 0.1\%$

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13. Study Procedures & Methodology

Management Pathway: Asciminib dosing schedule optimization (40 mg BID vs. 80 mg QD) **Education Protocol (HCP):** Site training on 2020 European LeukemiaNet (ELN) criteria for warning, failure, and intolerance **Education Protocol (Patient):** Oral chemotherapy adherence, dosing schedule compliance, and toxicity management **Quality Audit Mechanism:** Centralized BCR::ABL1 polymerase chain reaction (PCR) monitoring

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14. Observation & Follow-Up Schedule

Total Duration: 144 weeks of treatment + 4 weeks safety follow-up **Onsite Visit Cadence:** Baseline, Weeks 12, 24, 36, 48, 72, 96, 144 **Remote Monitoring Frequency:** Routine site-directed follow-ups **Checklist Review Interval:** Conducted at every scheduled efficacy/safety milestone

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | **____ | [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] | ___ |** ***[DD-MMM-YYYY]*** **| |** ***Lead Statistician*** **|** ***[Name]*** **|** _____ | [DD-MMM-YYYY] |

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16. Appendix: Supplementary Metadata & Database Mappings

SNOMED Hierarchy: 92812005|Chronic leukemia|Chronic leukemia, disease (disorder); 109993000|Chronic myeloproliferative disorder (clinical)|Chronic myeloproliferative disorder (clinical) (disorder); 188732008|Myeloid leukemia|Myeloid leukemia (disorder); 14016003|Bone marrow structure|Bone marrow structure (body structure); 1162588009|Chronic myeloid leukemia|Chronic myeloid leukemia (morphologic abnormality); 64572001|Disease|Disease (disorder)

Drug Reference (drug_name / drug_preferred_name): guaifenesin **ATC Code:** R05CA03 **Guaifenesin Trade Names:** Coldcough EXP, Relcof C, Liqufruta, Lortuss EX, VanaCof G, Cough Out, Dayquil Mucus Control, Kids-Eeze Chest Relief, Rompe Pecho, Codar GF, Certuss Cough, Diabetic Tussin Mucus Relief, Z-Tuss E, Refenesen, Diabetic Tussin EX, Bitex, Alphen, Cyndal, Endotuss, Biotussin, Cheratussin, Co-Histine, Decohistine, Deproist, Codahistine, Novagest, Tussar, Suttar, Phenhist, Tussi Organidin, Codegest, Naldecon CX, Deconsal C, Maxifed-G CD, Maxiphen CDX, ZTuss, Mar-cof CG, Coldcough XP, Brondil, Nucotuss, Vortex, Buckley's Chest Congestion, Exall, Donatuss DC, ExeClear-C, Tusso-ZR, Tusso ZMR, Humibid L.A. Reformulated Apr 2009, Poly Hist CB, Poly Tussin EX, Sudatuss 2 DF, Diabetic Choice, Duratuss CS, Allfen Reformulated Mar 2007, J-Max DHC, M-Clear WC, Nazarin HC, Allfen CDX, Allfen CD, Maxifed CD, Ambifed-G CD, Maxiflu CD, Maxiphen CD, Ambifed CD, Certuss, Qual-Tussin DC, Respi-Tann G, Tussicare DH, Monte-G HC, Tussiclear DH, Allfen CX, Allfen C, Humibid Maximum Strength, Elixsure Expectorant, Quala HC, Dynex VR, Cypex, Pulmari-GP, Uni-Guaifen 600/300, Dynatuss DF, Crantex HC, Dyphysin, Simuc-HD, A-Cof DH, Guaifen NR, Relasin-HCX, Allres G, Allfen Jr, Tusso C, COPD pharmacologic substance, Carbatuss Syrup, ZHist, Pseudatex HC, Humibid e, Dynex HD, Gentex HC, Atuss HX, Uni-Lev 5.0, Pancof XP Reformulated Feb 2008, Mintuss G, Condasin, Exetuss HC, ExeClear, Canges XP, Guiaplex HC, Narcof, Quintex HS, Nariz HC, Extendryl HC, Hydro-Tussin HG, Flutuss XP, Hydrofed, Execof XP, SoloGuai, MonteFlu HC, Z-Cof HCX, De-Chlor G, Dynatuss HC, Hydro-Tussin EXP, Pancof EXP, Pneumotussin, Uni-Cof Expectorant, Guaispan, Albatussin EX, Oratuss, Miltuss EX, Maxi Tuss HCG, Vita Numonyl Ex, Elixiophyllin GG, Tusso HC, Marcof, Sudatuss SF, Z Tuss, C Tussin, Gencofed Expectorant, GeneCof XP, Welltuss Exp, Tyrodone Expectorant, H-Quib, HYCOTUSS, Ambi, TUSS AX, Tusso DF, Vitussin, Carba XP, Genocidin Tuss, Betavent, Broncodur, XPECT, Su-Tuss HD, Touro HC, XPECT HC, Theospect, Dilex, Guaitussin AC, Despec EXP, GFN DYP, Guaitussin DAC, XPECT-AT, Neoasma, Liquidbid 1200, Tussafed-HC, Diabetic Tussin C, Drituss G, Entex HC Discontinued, Iofen, Guaifenesin DAC Liquid, Hydro-Tuss XP, Hydrotussin HD, Hydro GP, Muco-Fen, Mucinex, Orgadin, Dilaudid Cough, Levall 5.0, Guiadrine G-1200, Allfen, Breonesin, Guiatuss, Hytuss, Codafen, Altorant, Anti-Tuss, Asbron G, Aquamist, Atuss G, Bidex, Broncholate, Bronchial brand of guaifenesin-theophylline, Broncomar GG, Bron-Tuss, Brondelate, Bronkaid, Brontex, Chemdal Expectorant, Cheracol with Codeine, Cheratussin DAC, Codiclear DH, Co-Tussin, Co-Tuss V Expectorant, Codotuss, Cleartuss DH, Codafed, Cotuss-V, Cycofed Expectorant, D-G, Decotuss-HD, Deconamine CX, Diabetic Tus with Codeine, Dihistine, Dilex-G, Difil G, Dilor-G, Donatussin DC, Drituss HD, Duraganidin NR, Duratuss HD, Duratuss G, Dy-G, Dyfilin GG, Dyphyllin-GG, Dyflex-G, Dyline GG, Dyphyl GG, Efasin Expectorant SF, Ed Bron G, Endal Expectorant, Elixophyllin-GG, Enditussin Expectorant, Entuss D, Equibron G, Entuss, Extussive, Fentuss Expectorant, Fenex-LA, G-Tuss, Ganidin NR, G Bid, GG 200 NR, GG-Cen, Glytuss, Glyceryl T, Glydeine, Guaifenex LA, Guaiatussin DAC, Guaifen DAC, Gua-SR, Guiatuss DAC, Guaifen AC, Guaifenex G, Guai-Co, Guaiatussin, Guaifenesin LA, Guiatuss AC, Gua HC, Gua PC, Halotussin, Halotussin DAC, Halotussin AC, Humigen LA, Humibid L.A., Hycoclear, Humibid Pediatric, Hycoclar Tuss, Hycotuss Expectorant, Humavent LA, Hycosin Expectorant, Hydrotuss HD, Hydrocod-GF, Hytussin Expectorant, Hydrotuss, Hytuss 2X, Iodrol NR, Iofen-NF, KG-Fed Expectorant, KG-Tuss HD, Kwelcof, LGG brand of Dyphylline/Guaifenesin, Liquibid, Lotussin, M-Clear, Lufyllin-GG, Lufyllin-EPG, Mallotuss, Mastussin, Med-Hist Expectorant, Medcodin, Medtuss HD, Medent C, Mudrane GG-2, Mucobid-L.A., Mudrane GG, Naldecon-EX Senior, Mytussin, Mytussin AC, Mytussin DAC, Nalex Expectorant, Novadyne Expectorant, Nucodine Expectorant, Nucochem, Organidin NR, Pancof XP, Panfil, Phanasin, Phenylhistine Expectorant, Pneumomist, Poly Tussin XP, Primatene Dual Action, Propatuss Expectorant, Q-B, Quibron, Respa-GF, Robafen, Robafen AC, Robichem, Robichem AC, Romilar AC, Ryna CX, S-T Forte, SRC Expectorant, Scot-Tussin, Slo-Phyllin GG, Sinumist, Siltussin, Sorbutuss, Spantuss HD, Solu-Phyllin GG, Statuss, T-Tussin, Theocon, Theomar G.G., Theophyll-GG, Theolate, Theo G, Theocon Elixir, Touro EX, Tussadur-HD, Tusscidin, Tussend Expectorant, Uni Bronchial, Vanacon, Vanex Expectorant, Vi Q Tuss, Vicotuss, Vicodin Tuss, Bronkolixir, Novahistine Expectorant, Nucofed Expectorant, Primatene with ephedrine & guaifenesin, Ninjacof XG, Obredon, Quindal Solution, Flowtuss, Hycofenix, Zodryl DEC, Little Colds Mucus Relief, Broncomar Expectorant, Tab Tussin, Spec-tuss